

2. One-Shot Assessment of Model Weakness

From observed prediction loss to a confirmed weak region

Construct a partition, measure and confirm

1 Model observed loss

Calculate Brier loss and fit auxiliary error trees.

2 Construct the partition

Use weighted-leaf distance and fit PAM medoids.

3 Locate weakness

Assign observations and measure each region's loss.

4 Confirm the frozen rule

Freeze the selected region and evaluate fresh data.

The predictive model stays fixed throughout the assessment.

Use measured loss to assess regional weakness

MODEL BEING VALIDATED

Its predicted probability is compared with the observed outcome.

Brier loss = $(\text{probability} - \text{outcome})^2$.

These losses measure regional weakness.

AUXILIARY ERROR MODEL

Fit trees to the observed losses.

Use their leaf structure and fitted values to construct distances.

Those distances organize customer regions.

The error model supplies the representation; observed outcomes supply the regional loss measurements.

From observed loss to candidate regions

1 Freeze the predictor

40,000 discovery observations. Calculate each Brier loss: $(\text{outcome} - \text{probability})^2$.

2 Fit an error model

120 depth-two XGBoost trees predict those losses. The original predictor stays fixed.

3 Read leaves and values

Each observation reaches one leaf per tree. Keep both its identity and fitted contribution.

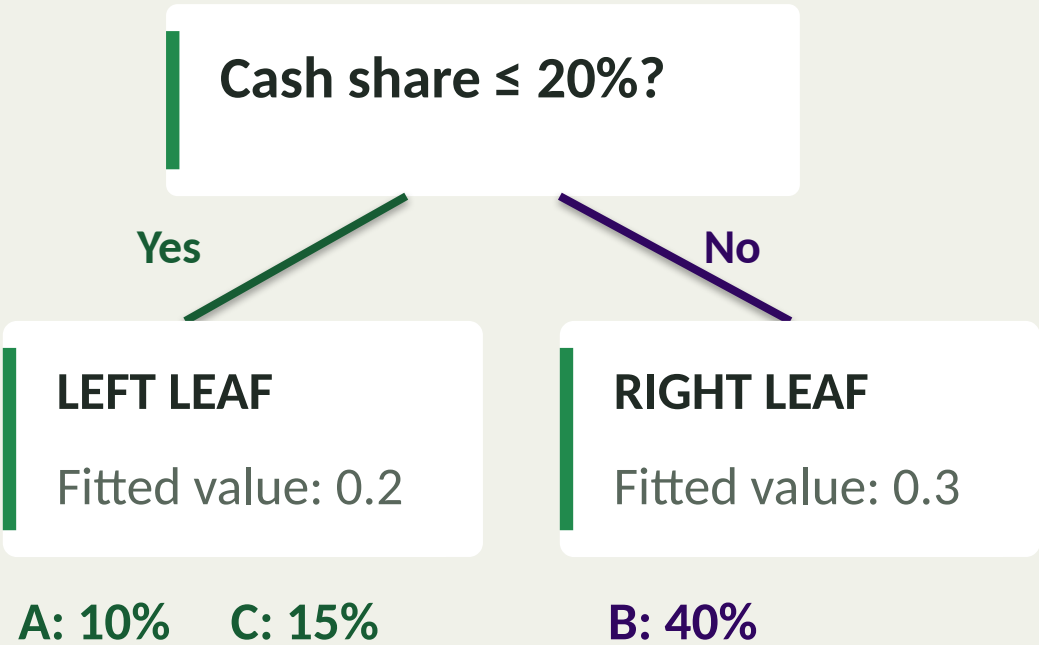
4 Construct and measure

Weighted leaf distance $\rightarrow$ four PAM medoids $\rightarrow$ observed loss in each cluster.

Example row: outcome 0, probability 0.018282, observed Brier loss 0.000334.

Each customer reaches one leaf

Route the customer through the tree, then record the value of the leaf reached.



One column for each leaf

Customer	Left	Right
A	0.2	0
B	0	0.3
C	0.2	0

Write the reached leaf's value in its column. Put zero in the other column.

Use the customer rows to calculate distance

A AND C REACH THE SAME LEAF

$$A = (0.2, 0)$$

$$C = (0.2, 0)$$

$$\text{Distance} = 0$$

$$\begin{aligned}\text{Shared-leaf proximity} &= 0.2 \times 0.2 \\ &= 0.04\end{aligned}$$

A AND B REACH DIFFERENT LEAVES

$$A = (0.2, 0)$$

$$B = (0, 0.3)$$

$$\begin{aligned}\text{Distance} &= \sqrt{[(0.2 - 0)^2 + (0 - 0.3)^2]} \\ &= 0.3606\end{aligned}$$

$$\text{Shared-leaf proximity} = 0$$

Combine leaf columns from all trees. Calculate one distance between complete customer rows.

Choose representatives for one partition

1 Supply the distances

Use the all-tree weighted-leaf distance matrix and choose K representatives.

2 Select starting medoids

Each representative is an observed customer.

3 Improve the representatives

PAM tests replacements that reduce total distance to the nearest medoid.

4 Return the fitted medoid set

Use the chosen representatives to assign all discovery observations.

PAM's fitting objective: minimize the sum of distances to the nearest representative.

Assign each customer to the nearest medoid

Inputs: the distance matrix and a given medoid set. Each medoid is an actual observation.

Example: $K = 2$; current medoids A and D

	A	B	C	D	E
A	0	1	4	9	10
B	1	0	3	8	9
C	4	3	0	5	6
D	9	8	5	0	1
E	10	9	6	1	0

Nearest medoid: {A, B, C} $\rightarrow$ A; {D, E} $\rightarrow$ D

1 Read the given medoids

Use columns A and D for this candidate.

2 Assign each observation

Join the medoid with the smallest distance.

3 Form the two clusters

Collect rows assigned to each medoid.

4 Measure model weakness

Calculate regional loss and contrast.

Keep the given medoids fixed throughout assignment and scoring.

Freeze the region before confirmation

DISCOVERY: SELECT A REGION

Assign all 40,000 observations.
Calculate mean Brier loss per cluster.
Require at least 50 rows and 1% share.
Select the highest-loss eligible cluster.

CONFIRMATION: APPLY THE RULE

Freeze trees, medoids and cluster ID.
Assign 200,000 fresh observations.
Measure regional loss and coverage.
Calculate contrast and uncertainty.

Use confirmation outcomes to measure the already-selected region.

Locate a concentrated region of model weakness

6.29% of observations

12,575 customers assigned to the frozen selected region.

0.09141 regional Brier loss

The pooled remaining customers have Brier loss 0.01589.

0.07552 loss contrast

Simultaneous 95% interval: 0.06963 to 0.08142.

27.85% of total Brier loss

Concentrated in this 6.29% population region.

Freeze trees, medoids and selected cluster; measure on a fresh sample.

Can we find a more concentrated weakness?

One fitted partition has exposed a high-loss region.
Its membership rule has been confirmed on a fresh sample.

Deck 3: construct several candidate medoid sets and use measured weakness to guide the next candidates.

Use discovery data for further search, then freeze the new selection for fresh confirmation.